# VS-04 Estimation, Analysis, and Statistical Modeling

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## VS-04.0 Estimation, Analysis, and Statistical Modeling

### Purpose and Scope

This document specifies the complete business layer for the **Estimation and closed-loop** business domain, corresponding to the *VS-04.0 Estimation, Analysis, and Statistical Modeling Value Stream*. It also covers 4 governance process types:

- **Design** — how the estimation, weighting, and analytical approach is specified and planned (produces metadata)
- **Implementation** — how the estimation and analysis pipeline is built, configured, and tested (produces the execution process)
- **Execution** — how data is operationally weighted, estimated, modeled, and validated (produces statistical data)
- **Management** — how the domain is monitored, quality-assessed, and improved (produces quality reports)

> **Out of Scope**
>
> Application and Technology layer specifications are out of scope for this document. Combining multiple analyses into integrated and composite statistical products is out of scope — it belongs to VS-05.0 Statistical Product Composition. Final disclosure control for publication and dissemination belongs to VS-06.0. VS-04 produces validated statistics with initial confidentiality protection; it does not compose products or publish them.

*A note on numbering:* each `VS-0x.y` below is both a value stream stage (*what value* is added) and a business process (*how* it is done); the numbering maps one-to-one. See the key-concepts box above.

### Position in the Statistical Value Chain

VS-04.0 is the fourth primary value stream — it receives Processed Data from VS-03.0 and applies statistical methods, estimation techniques, and analytical models to turn it into representative results that describe real-world phenomena. It is the "Estimate & Analyze Information" orchestration stage where data becomes statistical information.

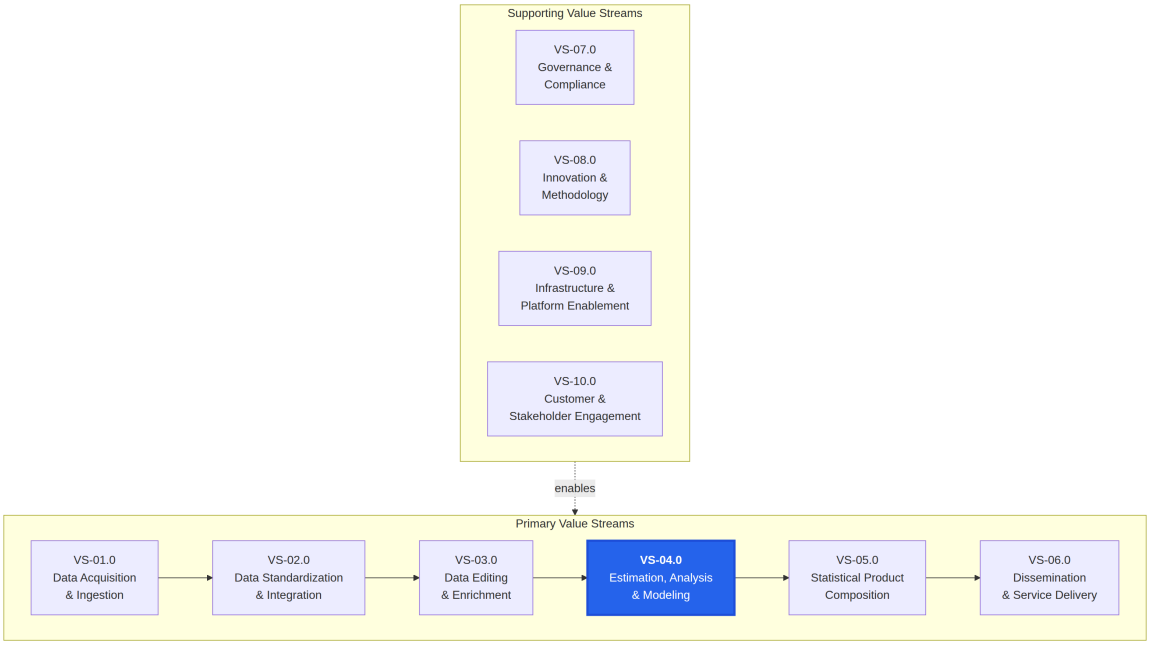



Figure 1: Value Stream Context Map. VS-04.0 is highlighted as the estimation-and-analysis stage that turns processed microdata into representative statistics.

### Steady State Transition

VS-04.0 transitions data from **Steady State #3 (Processed Data)** — edited, complete, enriched microdata — to **Steady State #4 (Output Data / Statistics)** — aggregated and analyzed statistics with estimates calculated, accuracy and quality indicators produced, lineage documented, internal review completed, and initial confidentiality measures applied so the output can potentially be released.

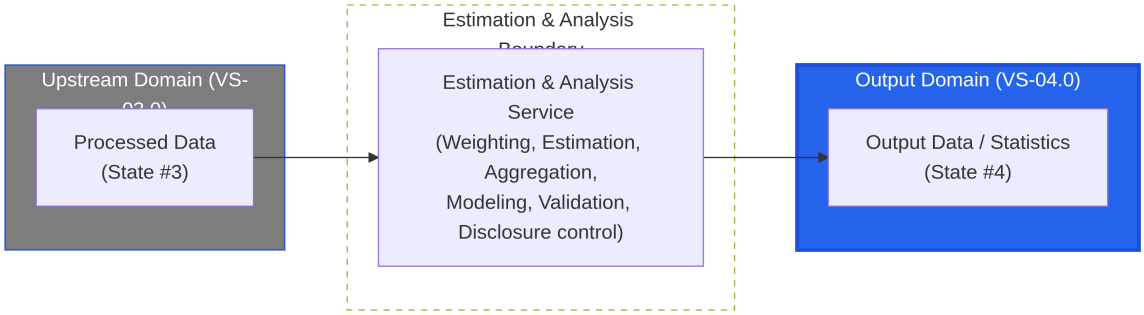



Figure 2: Steady State Transition from Processed Data to Output Data through the estimation and analysis service at the domain boundary. State #4 is shared with VS-05.0 Statistical Product Composition.

Output quality criteria (Examples)*

- Estimation/analytical approach selected and justified for the analytical objective
- Weighting and calibration applied to correct for sample bias, non-response, and coverage
- Estimates and aggregates calculated with accuracy measures (sampling error, confidence intervals)
- Model-based analyses conducted where applicable, with diagnostics documented
- Results validated for statistical soundness and coherence; methodology documented
- Initial statistical disclosure control applied so the output can potentially be released
- Full lineage from each statistic back to the processed source data and the method applied

The methodological discipline — insight, transparently produced

VS-04 does not change microdata values; it **computes statistics from them**. The output is a new kind of object — estimates, weights, aggregates, model results. The guarantee here is **methodological transparency, reproducibility, and statistical soundness**: every estimate carries the methodology and parameters that produced it, an accuracy measure (sampling error, confidence interval, or model diagnostic), and lineage to its inputs, so that any published figure can be regenerated and defended. Fitness for purpose and coherence with related statistics are explicit acceptance criteria.

Inherited from VS-03.0

VS-04 receives Processed Data **plus** the metadata VS-03 attached — edit lineage, derived-variable definitions, and **imputation flags**. The imputation flags matter here: estimation must reflect the additional uncertainty that imputed values introduce (e.g., via variance estimation that accounts for imputation), and the edit/derivation lineage is carried through so every estimate remains traceable to its original observed and imputed inputs.

### The Four-Process Governance Pattern

### Governing Principles

The following principles directly govern the VS-04.0 business domain:

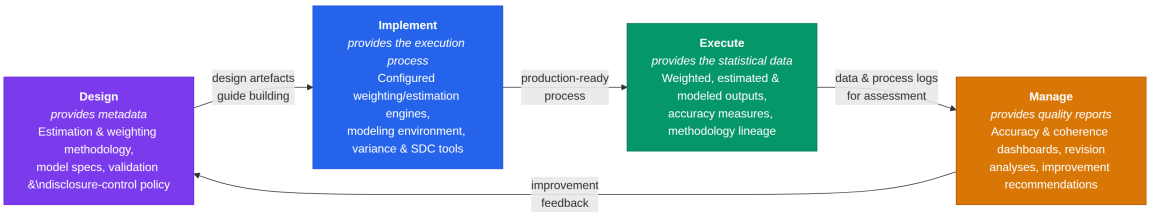



Figure 3: The Four-Process Governance Pattern. Each process type produces a distinct artefact class. The cycle closes when Management findings feed back into Design improvements.

| Principle | Statement | Application to VS-04.0 |
|---|---|---|
| **SAF-P01** | Actively participate in open source to maintain collective control over statistical processes | NSIs should contribute to open source communities for estimation, weighting, and modeling tooling (e.g., calibration and variance-estimation libraries), ensuring collective control over critical methodological capabilities rather than vendor dependence. |
| **SAF-P07** | Implement loosely coupled processes and systems | The estimation domain must be independent from upstream editing and downstream composition. Output Data (State #4) serves as a decoupling interface — downstream consumers access validated statistics without knowledge of the estimation internals. |
| **SAF-P08** | Data is shared property | Output statistics and output microdata, once validated, are available to all authorized consumers. Estimation should not be duplicated across products when the same validated output can serve multiple purposes. |
| **SAF-P09** | No data without metadata and classification | Every estimate carries metadata: the methodology and parameters applied, the weights used, accuracy measures (sampling error, confidence interval), the reference population, and a classification for derived statistics. This is what makes results transparent and reproducible. |
| **SAF-P10** | Security By Design | Access controls, integrity verification, and a tamper-evident record of methods and parameters are embedded in the estimation pipeline so that every published figure is attributable and reproducible, not added as an afterthought. |

| Principle | Statement | Application to VS-04.0 |
|---|---|---|
| **SAF-P11** | Privacy By Design | Initial statistical disclosure control is applied to outputs at this stage, and the disclosure risk of estimates and any output microdata is assessed by design, so the result can potentially be released. |

ℹ Cross-cutting principle application

Several principles cut across all four process types. SAF-P09 (no data without metadata and classification) is the backbone of methodological transparency: Design defines the methodology and required output metadata, Execute captures the method, parameters, weights, and accuracy measures for every estimate, and Manage assesses output quality and coherence. SAF-P10 (Security By Design) governs the tamper-evident record of methods and the reproducibility of outputs across Execution (Sections 5.1–5.5). SAF-P11 (Privacy By Design) applies especially to validation and disclosure control (Section 5.5), where the output's disclosure risk is assessed before handover. SAF-P07 (loosely coupled) governs both the State #3 intake interface from VS-03.0 and the State #4 handover interface to VS-05.0.

### Capability Foundations

VS-04.0 draws on the following SAF capabilities:

| Capability Group | Capabilities | Role in VS-04.0 |
|---|---|---|
| **1.1 Statistical design and build** | 1.1.2 Design statistics product, 1.1.3 Build statistics product | Design the estimation/weighting methodology and analytical models, and build the estimation pipeline |
| **2.1 Data management** | 2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management | Manage data, methodology metadata, and quality; assess accuracy and coherence; use registers for calibration totals |
| **3.2 Computerization and automation** | 3.2.1 Application development, 3.2.2 Functional management | Build and maintain weighting, estimation, aggregation, and modeling engines |
| **3.3 Security and Privacy** | 3.3.1 Privacy management, 3.3.2 Security planning | Apply statistical disclosure control and assess output disclosure risk |
| **4.1 Governance** | 4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improve management, 4.1.5 Accountability | Govern the domain, steward estimation methodology, plan improvements, report on reproducibility and compliance |

### Metadata Perspective

It is considered that metadata is an essential product of the Design phase and a critical concern of every other phase. In this domain, metadata is what makes a published figure defensible: every estimate must carry the method, parameters, weights, and accuracy that produced it. The table below consolidates the metadata produced across all four process types, covering both data metadata (about the outputs) and process metadata or paradata (about how the process runs).

| Process Type | Data Metadata Produced | Process Metadata (Paradata) Produced |
|---|---|---|
| **Design** | Estimation/weighting methodology, calibration totals & sources, aggregation specs, model specifications, validation & SDC policy | Design decisions log, methodology version, review and approval records |
| **Implement** | Engine configurations, weighting/estimation parameters, model parameters, variance-estimation settings, SDC parameters | Build log, test results, reproducibility-test outcomes, deployment approvals |
| **Execute** | Per-estimate methodology, weights, accuracy measures (sampling error, CI), model diagnostics, reference population, applied-SDC record | Operator identity, timing, software/version, parameter values, run identifiers |
| **Manage** | Accuracy & coherence assessments, revision analyses, lineage assessments, trend analyses | Accuracy dashboards, coherence reports, improvement decisions, audit findings |

## Business Roles and Actors

| Role | Process Orientation | Description |
|---|---|---|
| **Upstream Data Provider (VS-03.0)** | External (internal NSI) | Supplies Processed Data (State #3) with edit/imputation lineage to the estimation domain |
| **Methodology Authority** | External | Sets the estimation, weighting, and analytical methods the NSI conforms to (e.g., methodology unit, Eurostat guidance) |
| **Estimation Methodologist** | Design | Designs the estimation strategy, weighting/calibration methodology, and accuracy-measurement approach |
| **Model Designer** | Design | Specifies analytical and forecasting models, their inputs, assumptions, and diagnostics |
| **Disclosure-Control Designer** | Design | Designs the statistical disclosure control rules and disclosure-risk criteria for outputs |
| **Data Engineer** | Implementation | Builds and configures weighting, estimation, aggregation, and modeling engines and the pipeline |
| **System Configurator** | Implementation | Configures estimation services, calibration-total sources, and access controls |
| **Test Manager** | Implementation | Plans and executes testing of the estimation pipeline (technical, reproducibility, benchmark) |
| **Process Owner** | Management | Accountable for the end-to-end performance of the estimation and analysis domain |
| **Compliance Officer** | Management | Ensures reproducibility and disclosure-risk compliance, produces audit reports |
| **Quality Manager** | Management | Monitors accuracy/coherence KPIs, produces quality reports, and drives continuous improvement |
| **Survey Statistician** | Execution | Applies the designed weighting/calibration and computes estimates with accuracy measures |
| **Analyst / Modeler** | Execution | Conducts model-based analysis, regression, and forecasting, and documents diagnostics |
| **Subject-Matter Reviewer** | Execution | Reviews results for plausibility, coherence, and fitness for purpose, and signs off |

| Role | Process Orientation | Description |
|---|---|---|
| **Data Steward** | Execution | Registers output datasets and versions, manages metadata entries, and applies access controls |

## Design Processes — "provides metadata"

*Framework alignment: GSBPM Phases 1–2 — see Framework Mappings.*

The design phase produces the **metadata artefacts** that govern all subsequent phases — the estimation strategy, weighting/calibration methodology, aggregation and model specifications, and the validation and disclosure-control policy. Without well-designed methods, estimation becomes ad-hoc and indefensible, and management has no baseline to measure accuracy and coherence against.

### Design Estimation Strategy and Accuracy Approach

The starting point is choosing how the statistics will be estimated. This sub-process selects the estimation strategy appropriate to the source and analytical objective — design-based estimation for sample surveys, register-based aggregation, or model-assisted/model-based estimation for small domains — and defines the **accuracy-measurement** approach: how sampling error and confidence intervals are computed, including how the uncertainty introduced by VS-03 imputation is reflected in variance estimation. The team defines the target parameters, reference populations, and domains of estimation, aligned with GSBPM 2.5 (Design processing and analysis).

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Select estimation strategy per source and objective (design-based / register-based / model-based) | Estimation Strategy Document | GSBPM 2.5 |
| Define target parameters, reference populations, and domains of estimation | Estimation Specification | GSBPM 2.5 |
| Define accuracy-measurement approach (sampling error, CI), reflecting imputation uncertainty | Accuracy Measurement Plan | GSBPM 2.5 |
| Define coherence criteria with related and prior statistics | Coherence Criteria | GAMSO QM |

**NSI Example — Labour Force Survey:**

The LFS team designs a design-based estimation strategy for the employment rate, with calibration to population totals by age, sex, and region, and a variance estimator (e.g., a linearization or replication method) that accounts for the survey design and for income/non-response imputation flagged in VS-03. Target parameters and the domains of estimation (national and NUTS-2) are specified.

### Design Weighting and Calibration Methodology

This sub-process designs how sample bias, non-response, and coverage errors are corrected through weighting. The team specifies the design weights, the non-response adjustment model, and the calibration method (e.g., calibration to known population totals from registers), including the auxiliary variables and calibration totals to be used and their authoritative sources. Bounds and acceptance constraints on weights are defined to prevent extreme weights, aligned with GSBPM 2.5.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Specify design weights and non-response adjustment | Weighting Specification | GSBPM 2.5 |
| Specify calibration method, auxiliary variables, and calibration totals (with sources) | Calibration Specification | GSBPM 2.5 |
| Define weight bounds and acceptance constraints | Weight Constraint Policy | GSBPM 2.5 |
| Define the authoritative source and version for calibration totals | Calibration-Total Source Register | CSDA Reference Data |

**NSI Example — Household Survey:**

The household survey calibration is designed to population totals by age × sex × region drawn from the population register, with a generalized regression (GREG) calibration and weight bounds to cap the calibration factor, so that weighted demographic margins reproduce the register while extreme weights are avoided.

### Design Aggregation and Model Specifications

This sub-process specifies what is computed. For aggregation, each output statistic is defined by its formula, the domains/breakdowns, and rounding/units conventions. For analysis, model-based methods (regression, time-series/forecasting, small-area models) are specified with their inputs, assumptions, and the diagnostics that must be produced and checked. Each output statistic is given its metadata and classification per SAF-P09, aligned with GSBPM 2.5.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Specify output aggregates: formulae, domains/breakdowns, units, rounding | Aggregation Specification | GSBPM 2.5 |
| Specify analytical/forecasting models: inputs, assumptions, diagnostics | Model Specification | GSBPM 2.5 |
| Define metadata and classification for every output statistic | Output-Statistic Metadata Standard | SAF-P09 |
| Define seasonal adjustment / time-series treatment where applicable | Time-Series Treatment Spec | GSBPM 2.5 |

### Design Validation and Disclosure-Control Policy

This sub-process designs how results are validated and protected before handover. Validation criteria cover statistical soundness (plausibility, consistency, model diagnostics within bounds), coherence with related and prior statistics, and the revision policy. The **statistical disclosure control (SDC)** policy is designed — disclosure-risk criteria for tables and any output microdata, and the protection methods (suppression, rounding, perturbation) — so the output reaches a state where it can potentially be released, aligned with GSBPM 6 design and SAF-P11. Reproducibility is designed in: an output must be regenerable from the processed input plus the recorded methodology and parameters.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Define validation criteria: soundness, coherence, model diagnostics, revision policy | Validation Policy | GAMSO QM |
| Define disclosure-risk criteria and SDC protection methods | Disclosure-Control Policy | SAF-P11 |
| Define output metadata and methodology-documentation requirements | Output Documentation Standard | SAF-P09 |
| Design reproducibility: regenerate outputs from inputs + methodology + parameters | Reproducibility Design | SAF-P10 |

#### Design Quality-Measurement and Versioning Framework

The final design sub-process ensures accuracy is measured and outputs are traceable. It defines the quality indicators to be produced (sampling error, coefficient of variation, response/coverage rates, model fit), the **output-versioning** policy (how versions are identified, what triggers a revision, immutability and retention), the lineage requirements from each statistic back to its processed inputs and the method applied, and the role-based access controls for outputs in line with SAF-P10.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Define quality indicators (sampling error, CV, response/coverage, model fit) | Quality Indicator Set | GAMSO QM |
| Define output-versioning and revision policy | Versioning Policy | CSDA Traceability |
| Define lineage from each statistic to processed inputs and method | Lineage Specification | CSDA Traceability |
| Design role-based access control for outputs | Access Control Design | SAF-P10 |

## Implementation Processes — "provides the execution process"

*Framework alignment: GSBPM Phase 3 (Build) — see Framework Mappings.*

The implementation phase takes the metadata artefacts from Design and **builds, configures, and tests** the production-ready estimation and analysis pipeline. Nothing runs operationally until this phase is complete.

#### Build Weighting and Calibration Engines

This sub-process turns the weighting and calibration designs into operational tools. The weighting engine is built or configured to compute design weights and non-response adjustments, and the calibration engine to calibrate to the designed totals using the specified auxiliary variables, enforcing the weight bounds. The calibration-total connections are configured to the authoritative register/reference sources with version pinning, so the totals used are recorded and reproducible. All weighting logic is version-controlled in line with SAF-P10, following GSBPM 3.1–3.2.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Build/configure the weighting engine (design weights, non-response adjustment) | Configured weighting engine | GSBPM 3.2 |

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Build/configure the calibration engine with auxiliary variables and bounds | Configured calibration engine | GSBPM 3.2 |
| Configure calibration-total connections (versioned register sources) | Calibration-total integration | CSDA Reference Data |
| Version-control all weighting/calibration logic | Versioned methodology repository | SAF-P10 |

**NSI Example — Household Survey:**

The team configures a GREG calibration engine that pulls age × sex × region totals from a pinned version of the population register, applies weight bounds, and records the calibration totals and convergence diagnostics with every run so the weighting is reproducible.

#### Build Estimation and Aggregation Pipelines

The designed estimation and aggregation specifications are implemented as automated services. The estimation engine computes the target parameters with the designed accuracy measures, and the **variance-estimation** component computes sampling errors and confidence intervals — accounting for the survey design and for imputation uncertainty using the VS-03 imputation flags. The aggregation pipeline computes the specified output aggregates across the designed domains with the rounding/units conventions. Every output is produced with its methodology, parameters, weights, and accuracy measure attached, following GSBPM 3.2.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Build the estimation engine for the designed target parameters | Estimation pipeline | GSBPM 3.2 |
| Build the variance-estimation component (design + imputation uncertainty) | Variance/accuracy component | GSBPM 3.2 |
| Build the aggregation pipeline (domains, units, rounding) | Aggregation pipeline | GSBPM 3.2 |
| Attach methodology, weights, and accuracy measure to every output | Output metadata binding | SAF-P09 |

#### Build Modeling Environment, Validation, and SDC Tooling

This sub-process implements model-based analysis, validation, and disclosure control. The modeling environment is built/configured to run the specified analytical, forecasting, and small-area models with their diagnostics, and (where applicable) seasonal adjustment. Validation tooling implements the designed soundness and coherence checks, including comparison against prior periods and related statistics. The **SDC tooling** implements the designed disclosure-risk checks and protection methods (suppression, rounding, perturbation). The output-versioning store is built so each accepted output version is immutable, identified, reproducible from inputs plus methodology, and accompanied by its lineage. Event notifications are configured to alert downstream consumers (VS-05.0) when an output version is registered.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Build/configure the modeling environment (models, diagnostics, seasonal adjustment) | Modeling environment | GSBPM 3.2 |

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Implement validation tooling (soundness, coherence, prior-period comparison) | Validation tooling | GSBPM 3.2 |
| Implement SDC tooling (disclosure-risk checks, suppression/rounding/perturbation) | SDC tooling | SAF-P11 |
| Build the output-versioning store (immutable, identified, reproducible) | Versioning store | CSDA Traceability |
| Configure "Output Data Registered" event for downstream consumers | Event publication | GSBPM 3.2 |

#### Test and Validate the Pipeline

Before the pipeline goes into production, it must be rigorously verified. Technical testing covers all individual components — weighting, calibration, estimation, variance, aggregation, modeling, validation, SDC, and versioning store (GSBPM 3.5). Integration testing validates the end-to-end flow from processed intake through weighting, estimation, modeling, validation, SDC, to versioned registration. A **reproducibility test** confirms an output can be regenerated identically from the processed input plus the recorded methodology and parameters; a **benchmark/back-series test** compares results against a known or prior-period benchmark; and accuracy measures are checked against expectations. The phase concludes with finalization of the production system and documentation of operational procedures (GSBPM 3.6–3.7).

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Technical testing of all individual components | Unit test reports | GSBPM 3.5 |
| Integration testing: end-to-end flow from processed intake to versioned registration | Integration test report | GSBPM 3.5 |
| Reproducibility test (regenerate an output from inputs + methodology) | Reproducibility report | GSBPM 3.6 |
| Benchmark / back-series test against known or prior results | Benchmark test report | GSBPM 3.6 |
| Check accuracy measures and model diagnostics against expectations | Accuracy verification report | GSBPM 3.6 |
| Finalize production system and document operational procedures | Operational Procedures Manual, Production-Ready Pipeline | GSBPM 3.7 |

## Execution Processes — "provides the statistical data"

*Framework alignment: the later GSBPM phases for this domain — see Framework Mappings.*

The execution phase runs the **designed and implemented pipeline** operationally for each production cycle. This corresponds to the five VS-04 value stream stages. In a mature steady state, much of the work has been pre-designed (Section 3) and pre-built (Section 4) — execution focuses on running the cycle, computing weights/estimates/models, and capturing the methodology, parameters, and accuracy measures so that every output is reproducible and defensible.

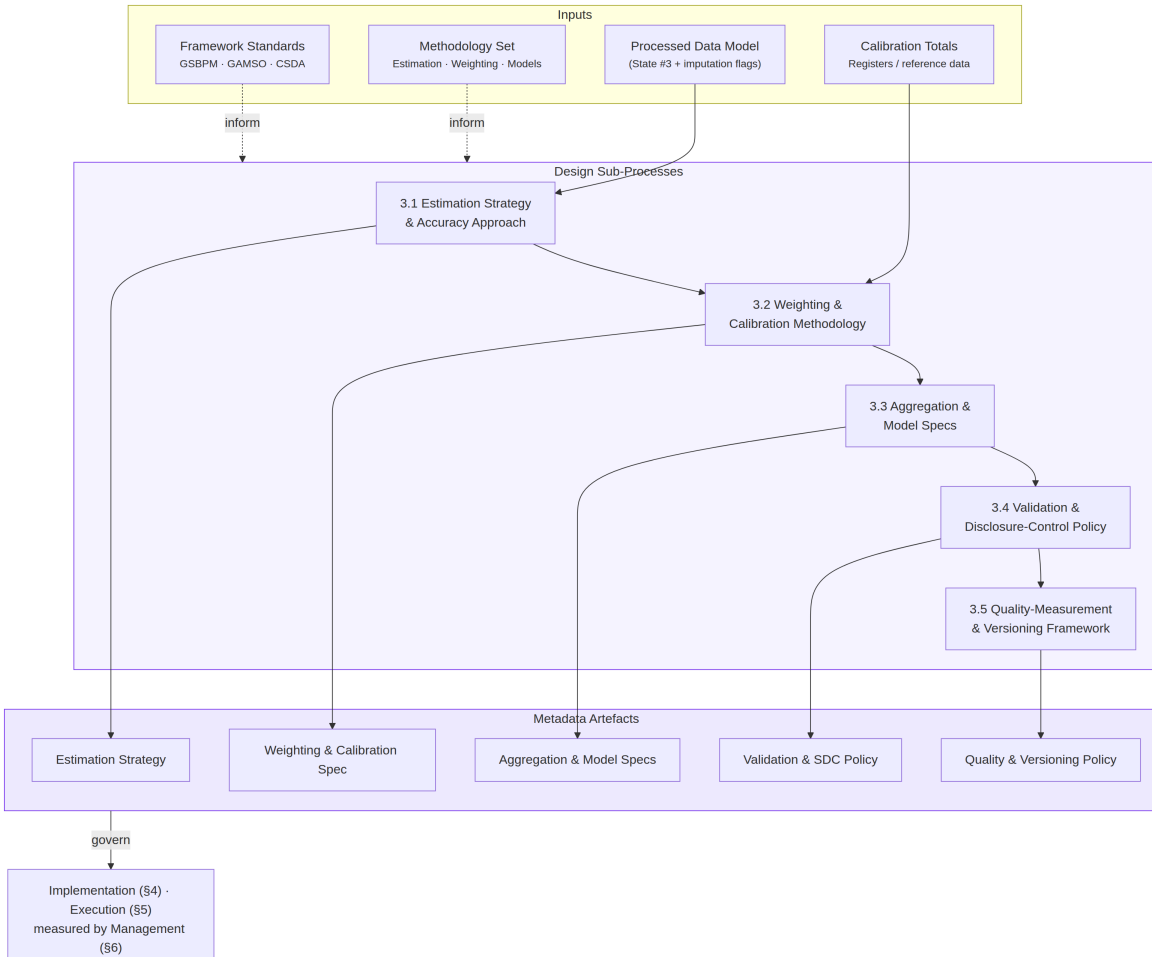



Figure 4: Design Process Decomposition. The processed data model, methodology set, calibration totals, and framework standards flow into five design sub-processes, producing metadata artefacts that govern Implementation, Execution, and Management.

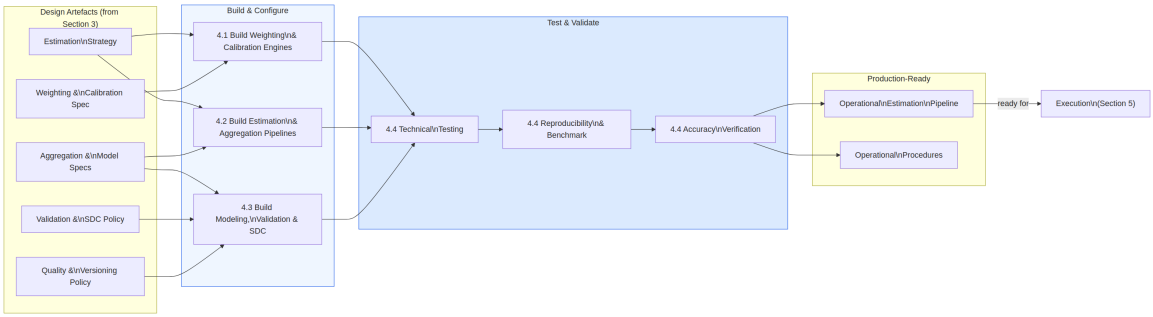



Figure 5: Implementation Pipeline from design artefacts through build/configure/test to production-ready system.

### Confirm Estimation and Analytical Approach

**Value created:** *The right, pre-approved method selected for this cycle's data and objective.*

When a "Processed Data Registered" event is received from VS-03.0, execution begins by confirming the estimation and analytical approach for the cycle. In a mature steady state the methodology has been designed (Section 3), so per-cycle execution confirms or selects the pre-approved approach for the data at hand, verifies that the calibration totals and reference sources are current, and confirms that the target parameters and domains apply. Significant changes (a new domain, a methodology revision) trigger a return to the design process rather than ad-hoc adaptation. This applies GSBPM 2.5/5.6 in their recurrent form.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Confirm or select the pre-approved estimation/analytical approach for the cycle | Confirmed approach for cycle | GSBPM 2.5 |
| Verify calibration totals and reference sources are current | Reference-source confirmation | CSDA Reference Data |
| Confirm target parameters and domains of estimation | Confirmed estimation scope | GSBPM 5.6 |
| Flag methodology changes for design review | Methodology-change exception (if any) | GAMSO QM |

**NSI Example — Labour Force Survey:**

For cycle Q3-2026 the LFS team confirms the design-based estimation strategy and the calibration totals (latest population-register extract by age × sex × region), and confirms the domains (national and NUTS-2). No methodology change is needed, so the cycle proceeds without a return to design.

### Apply Weighting and Calibration

**Value created:** *Bias from sampling, non-response, and coverage corrected so results represent the population.*

This sub-process computes the weights, aligned with GSBPM 5.6 (Calculate weights). The weighting engine applies the design weights and non-response adjustment, and the calibration engine calibrates to the confirmed population totals using the designed auxiliary variables, enforcing the weight bounds. Calibration convergence and the resulting weight distribution are checked against the designed constraints, and the calibration totals, auxiliary variables, and convergence diagnostics are recorded so the weighting is reproducible.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Compute design weights and non-response adjustment | Adjusted weights | GSBPM 5.6 |
| Calibrate to confirmed population totals with designed auxiliary variables | Calibrated weights | GSBPM 5.6 |
| Check convergence and weight distribution against bounds | Weight diagnostics | GSBPM 5.6 |
| Record calibration totals, auxiliaries, and diagnostics as lineage | Weighting lineage | SAF-P09 |

**NSI Example — Household Survey:**

The GREG calibration converges in 6 iterations; weighted margins reproduce the register totals by age × sex × region, the calibration factor stays within the designed bounds, and the totals and diagnostics are stored with the run.

### Estimate and Aggregate Data

**Value created:** *Representative statistics and aggregates, each carrying its accuracy.*

This sub-process computes the target estimates and aggregates, aligned with GSBPM 5.7 (Calculate aggregates) and 6.1 (Prepare draft outputs). The estimation engine computes the target parameters using the calibrated weights, and the aggregation pipeline computes the specified aggregates across the designed domains with the units/rounding conventions. The variance-estimation component computes sampling errors and confidence intervals, reflecting both the survey design and the imputation uncertainty signaled by the VS-03 imputation flags. Every estimate is produced with its methodology, weights, and accuracy measure attached.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Compute target estimates using calibrated weights | Point estimates | GSBPM 5.7 |
| Compute aggregates across designed domains (units, rounding) | Output aggregates | GSBPM 5.7 |
| Compute sampling errors and CIs (design + imputation uncertainty) | Accuracy measures | GSBPM 6.1 |
| Attach methodology, weights, and accuracy to every estimate | Estimate metadata | SAF-P09 |

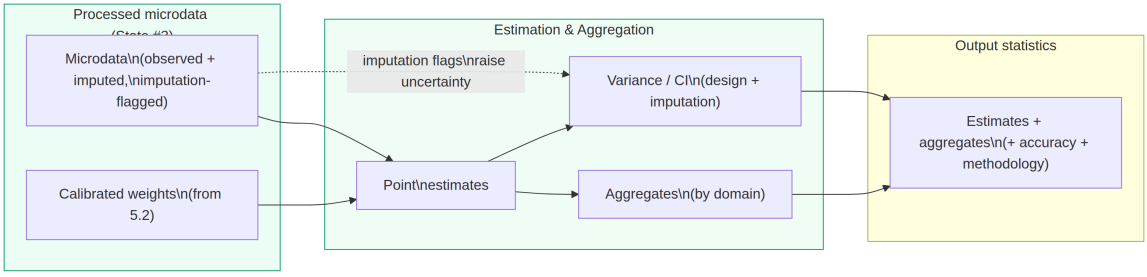



Figure 6: Estimation and Aggregation Architecture. Calibrated weights and imputation-flagged microdata feed point estimation and aggregation; variance estimation reflects both the survey design and the imputation uncertainty, and every output carries its accuracy and methodology.

**NSI Example — Labour Force Survey:**

The Q3-2026 employment rate is estimated at 71.4% with a 95% CI of ±0.5pp at national level and wider intervals at NUTS-2; the variance estimate is inflated appropriately for the income/non-response imputation flagged in VS-03. Each published cell carries its estimate, CV, and the method that produced it.

### Perform Model-Based Analysis

**Value created:** *Deeper insight — relationships, trends, and small-domain estimates the direct counts cannot give.*

This sub-process conducts the designed model-based analyses, aligned with GSBPM 6.1/6.3. Analytical, regression, time-series/forecasting, and small-area models specified in design are run on the data, their diagnostics computed and checked against the designed bounds, and seasonal adjustment applied where applicable. Results are interpreted in context, with assumptions and limitations documented. As with direct estimates, every model output carries its specification, parameters, diagnostics, and uncertainty so it is reproducible and defensible.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Run the designed analytical/forecasting/small-area models | Model outputs | GSBPM 6.1 |
| Compute and check model diagnostics against designed bounds | Model diagnostics | GSBPM 6.2 |
| Apply seasonal adjustment / time-series treatment where applicable | Adjusted series | GSBPM 6.1 |
| Interpret results; document assumptions and limitations | Analytical commentary | GSBPM 6.3 |

**NSI Example — Tax Register to Business Statistics:**

For small NACE × region domains where the direct register aggregate is unstable, a small-area model borrows strength across domains to produce more reliable estimates; model diagnostics (fit, prediction intervals) are recorded, and the model-based cells are flagged distinctly from direct aggregates.

#### Validate and Document Results

**Value created:** *Statistically sound, coherent, documented, and disclosure-safe statistics ready for handover.*

The final execution sub-process validates the results and produces the official output version, aligned with GSBPM 6.2 (Validate outputs), 6.4 (Apply disclosure control), and 6.5 (Finalise outputs). Results are checked for statistical soundness and consistency, and for coherence with prior periods and related statistics; significant movements are explained, and revisions handled per policy. **Initial statistical disclosure control** is applied per the designed policy — disclosure risk assessed and protection methods (suppression, rounding, perturbation) applied — so the output can potentially be released. The methodology and quality indicators are documented, a persistent immutable output version is created with a unique identifier and full lineage, registered per SAF-P09, and access permissions applied per SAF-P10. The sub-process concludes by publishing an "Output Data Registered" event, notifying downstream VS-05.0 that new validated statistics are available for composition.

**Output: Output Data / Statistics (Steady State #4)** — the official handover artefact to the rest of the statistical value chain.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Validate soundness and coherence; explain movements; handle revisions | Validation Report | GSBPM 6.2 |
| Apply initial statistical disclosure control per designed policy | Disclosure-controlled outputs | GSBPM 6.4 |
| Document methodology and quality indicators | Methodology & Quality Documentation | SAF-P09 |
| Create an immutable, identified output version with full lineage | Versioned Output Data | CSDA Traceability |

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Register metadata and publish "Output Data Registered" event to notify VS-05.0 | Catalog registration, event | SAF-P07 |

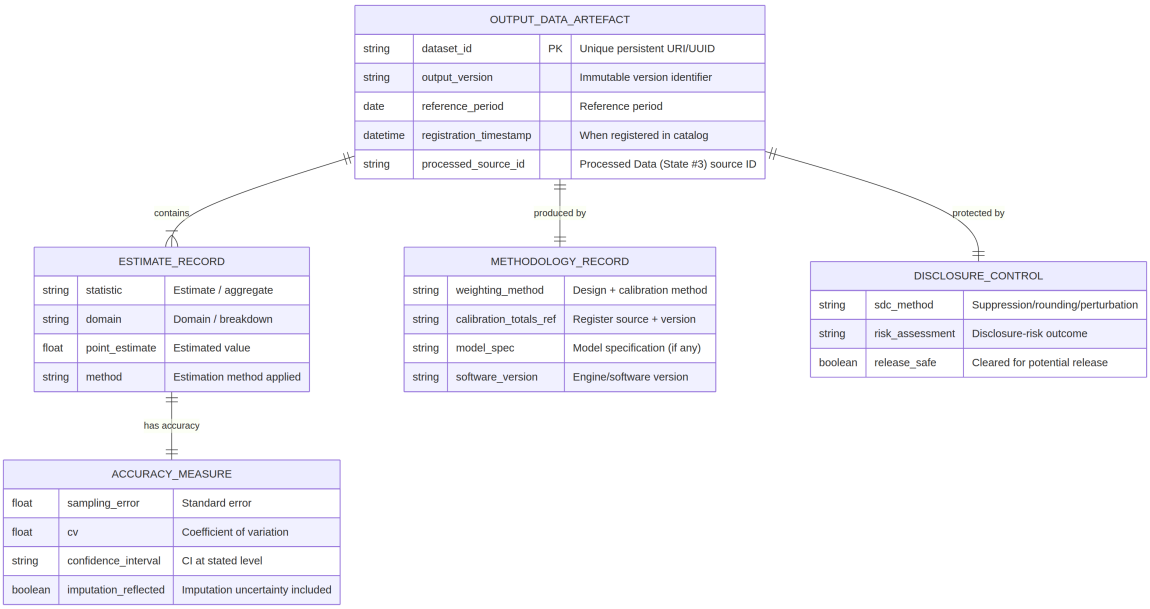



Figure 7: Output Data Artefact Structure. Reading guide: each box is a type of record with its fields; a connecting line means the records are linked. Each output version contains estimates with accuracy measures, is linked to the methodology that produced it and the applied disclosure control, and is traceable to the processed source data.

## Management Processes — "provides quality reports"

*Framework alignment: GSBPM overarching Quality/Metadata Management — see Framework Mappings.*

The management phase runs **alongside and after** execution. It monitors the estimation domain's performance, assesses the accuracy and coherence of outputs, ensures every result is reproducible and disclosure-safe, and feeds improvement recommendations back into the design phase — closing the governance cycle.

### Monitor Process Performance

Operational performance of the estimation domain must be tracked systematically across production cycles. The process owner and quality manager collect and aggregate key performance indicators, monitor throughput against the release calendar, and track paradata — which methods and parameter/software versions were run, when, by whom, and how many manual interventions occurred. This information feeds process performance dashboards that give the process owner a real-time view of domain health, aligned with the GAMSO Quality Management capability.

**Key Performance Indicators (KPIs):**

| KPI | Description | Measurement |
|---|---|---|
| On-time output rate | Outputs produced within the release-calendar window | Per output, per cycle |
| Average sampling error / CV | Typical accuracy of key estimates | Per statistic, per cycle |

| KPI | Description | Measurement |
|---|---|---|
| Coherence-check pass rate | Outputs passing coherence checks on first run | Per cycle |
| Revision magnitude | Size of revisions vs. previously published figures | Per statistic, per cycle |
| Reproducibility rate | Outputs regenerable identically from inputs + methodology | Per cycle |
| Processing time | Duration from processed intake to versioned registration | Per output, per cycle |

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Collect and aggregate KPIs per production cycle | Process Performance Dashboard | GAMSO QM |
| Monitor throughput against the release calendar | Throughput Report | GAMSO QM |
| Track paradata: methods, parameter/software versions, interventions | Paradata Records | GAMSO QM |
| Generate process performance dashboards | Dashboard for Process Owner | GAMSO QM |

#### Assess Output Quality

Beyond process performance, the quality of the outputs themselves must be evaluated against the criteria established in design. Each output is assessed across multiple quality dimensions: accuracy (sampling error, CV, model fit), coherence (consistency with related and prior statistics), timeliness (against the release calendar), and reliability (revision analysis comparing preliminary and final figures). Results are compared against the designed quality criteria from Section 3.5. Quality indicators are produced per statistic, per output, and per cycle, and trends are tracked to detect deteriorating accuracy, growing revisions, or models that need recalibration. This aligns with the CSDA Data Quality capability and SAF Capability 2.1.3.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Evaluate accuracy (sampling error, CV, model fit) | Accuracy Report (per cycle) | SAF 2.1.3 |
| Assess coherence with related and prior statistics | Coherence Report | CSDA Data Quality |
| Conduct revision analysis (preliminary vs. final) | Revision Analysis | SAF 2.1.3 |
| Identify quality trends across statistics and cycles | Quality Trend Analysis | SAF 2.1.3 |

#### Manage Methodology and Output-Version Lifecycle

The estimation methodology and the output versions it produces must remain current, complete, and well-governed. This sub-process governs the lifecycle of estimation methods, weighting/calibration specifications, and models — versioning them, managing transitions (e.g., a methodology change with back-series implications), and retiring obsolete methods. It verifies metadata completeness — that every output carries full methodology, weights, accuracy measures, and lineage as an SAF-P09 compliance check — and governs the output-version store: retention, immutability, and reconciliation that catalog entries match stored versions. Metadata lineage from processed through estimation to output state is tracked.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Govern estimation-method, weighting, and model lifecycle and versions | Methodology Version Register | SAF-P09 |
| Verify metadata completeness (methodology, weights, accuracy, lineage) | Metadata Quality Report | SAF-P09 |
| Govern the output-version store: retention, immutability, reconciliation | Version Store Audit | CSDA Traceability |
| Track metadata lineage from processed to output state | Metadata Lineage Audit | CSDA Traceability |

#### Audit and Compliance Reporting

This sub-process produces the evidence that the estimation domain operates within methodological, legal, and organizational requirements. Reproducibility-audit reports demonstrate that every published figure can be regenerated from the processed input plus the recorded methodology and parameters — supporting methodological transparency and accountability. Disclosure-control reporting confirms that SDC was applied and that outputs meet the disclosure-risk criteria per SAF-P11. Methodology compliance reporting confirms that approved estimation and analytical methods were applied, and security compliance reporting verifies that designed access controls and the tamper-evident methodology record have been applied. These reports satisfy the SAF Capability 4.1.5 (Accountability) obligation and feed into VS-07.0 Governance and Compliance.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Produce reproducibility-audit reports: every figure regenerable | Reproducibility Audit Summary | SAF 4.1.5 |
| Report on disclosure control applied and disclosure-risk outcomes | Disclosure-Control Report | SAF-P11 |
| Report on methodology compliance (approved methods applied) | Methodology Compliance Report | SAF 4.1.5 |
| Report on security compliance: access controls, methodology-record integrity | Compliance Report (per cycle) | SAF-P10 |

#### Continuous Improvement

The final management sub-process closes the governance loop by analyzing performance and quality data to identify and drive improvements. Quality trends across production cycles are examined: are sampling errors falling, are revisions shrinking, is coherence improving, are models holding up? Improvement opportunities are identified — refining the calibration specification to reduce variance, recalibrating or replacing models, improving non-response adjustment, adopting better variance estimators, and tuning the release process to improve timeliness. These improvement recommendations are fed back to the design processes (Section 3), prioritized based on impact and feasibility, and tracked through subsequent design-implement-execute cycles. This is the SAF Capability 4.1.4 (Improve management) in action, and it connects to the GAMSO concept of transferring mature capabilities from innovation into production.

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Analyze quality trends across production cycles | Improvement Recommendations Report | SAF 4.1.4 |

| Activity | Key Outputs | Framework Ref |
|---|---|---|
| Identify improvement opportunities: calibration, models, variance estimators, timeliness | Updated Design Inputs | SAF 4.1.4 |
| Feed recommendations back to design processes (Section 3) | Design feedback (closing the governance loop) | SAF 4.1.4 |
| Prioritize improvements based on impact and feasibility | Improvement Tracking Log | GAMSO Capability Mgmt |
| Track implementation of accepted improvements | Improvement Tracking Log | GAMSO Capability Mgmt |

## Business Services and Interfaces

The VS-04.0 business domain exposes the following services:

| Service | Direction | Consumers | Mapped Processes |
|---|---|---|---|
| **Weighting and Calibration Service** | Internal (VS-03.0 → NSI) | Processed data consumers, estimation teams | 5.2 Apply Weighting and Calibration |
| **Estimation and Aggregation Service** | Internal | Estimation teams, analysts | 5.3 Estimate and Aggregate Data |
| **Modeling and Analysis Service** | Internal | Analysts, methodologists | 5.4 Perform Model-Based Analysis |
| **Validation and Disclosure-Control Service** | Internal | Quality managers, disclosure-control officers | 5.5 Validate and Document Results |
| **Output-Data Provisioning Service** | Internal (NSI → VS-05.0) | Product composition teams, data analysts | 5.5 Validate and Document Results |
| **Output-Version Catalog Service** | Internal | All data consumers across the NSI | 6.3 Manage Methodology & Version Lifecycle |
| **Design and Methodology Service** | Internal | Methodologists, estimation/model designers | 3.1–3.5 All design processes |

**Interface specifications:**

- **Output-Data Provisioning Service** interface: event-driven notification ("Output Data Registered") + catalog query API for downstream consumers to discover and access output versions, their accuracy measures, and methodology
- **Estimation and Aggregation Service** interface: estimation API returning estimates with accuracy measures, and a methodology/parameters query API
- **Output-Version Catalog Service** interface: search/browse API for output versions, accuracy measures, methodology documentation, and lineage

### Boundary Interface Specifications

The domain is loosely coupled (SAF-P07) to its neighbors through two steady-state interfaces. Specifying each boundary explicitly — the handover artefact, the metadata it must carry, the event that triggers it, and where responsibility transfers — makes the loose-coupling contract concrete and testable.

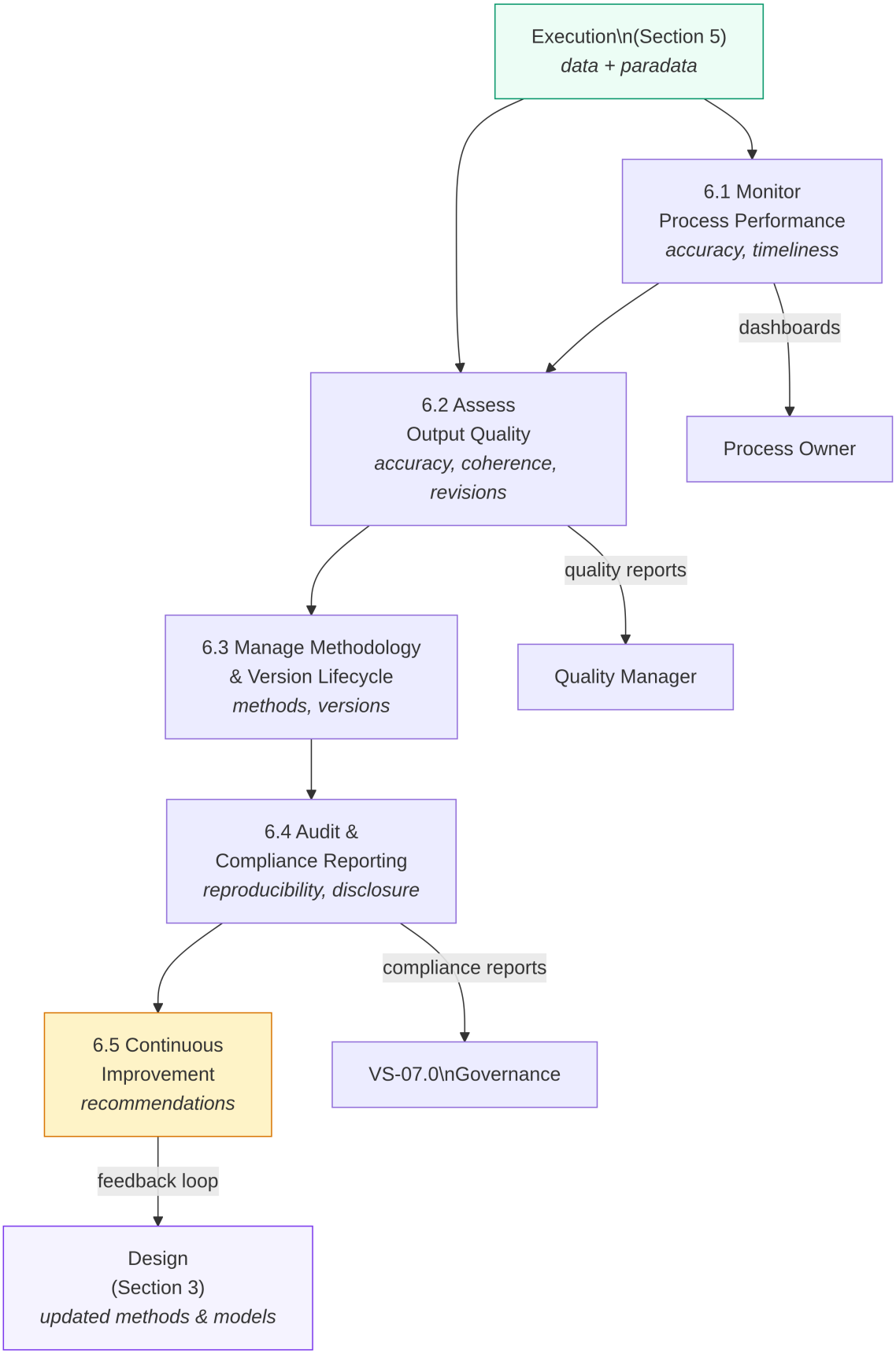



Figure 8: Management Feedback Loop. Execution data flows through monitoring, quality assessment, methodology/version management, and audit processes. Continuous improvement feeds recommendations back to Design, closing the governance cycle.

| Boundary | Handover Artefact | Required Metadata | Trigger Event | Responsibility Transfer |
|---|---|---|---|---|
| **VS-03.0 → VS-04.0** (intake) | Processed Data (State #3) | Edit lineage, derived-variable definitions, **imputation flags and methods**, quality-improvement assessment, quality-version identifier, BIV classification | "Processed Data Registered" event | VS-03.0 retains custody of the processed version; VS-04.0 assumes responsibility for weighting, estimation, modeling, and validation on event receipt |
| **VS-04.0 → VS-05.0** (handover) | Output Data / Statistics (State #4) | Methodology and parameters, weights, accuracy measures (sampling error, CI), model diagnostics, applied-SDC record, quality indicators, output-version identifier, lineage to processed source, BIV classification | "Output Data Registered" event | VS-04.0 retains custody of the output version; VS-05.0 assumes responsibility for composing integrated and composite products on event receipt |

Each interface exchanges a registered, immutable steady-state artefact plus its metadata — never internal working data — so neither side needs knowledge of the other's internal processing.

## Business Objects and Data Flow

## Business Events

## Framework Mappings

This reference table maps each process group of this document to the international frameworks (GSBPM, GAMSO, CSDA) and the SAF capability model. It is intended for architects and auditors; the narrative sections above do not depend on it.

| Process group | GSBPM | GAMSO | CSDA | SAF Capabilities |
|---|---|---|---|---|
| **Design** | Phase 1 (Specify Needs), Phase 2 (Design: 2.1–2.5) | Production (design activities), Capability Management | Data Governance, Metadata Management, Data Quality | 1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management |
| **Implementation** | Phase 3 (Build: 3.1–3.7) | Capability Management | Data Processing / Analysis (setup), Data Governance (configuration) | 1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management |

| Process group | GSBPM | GAMSO | CSDA | SAF Capabilities |
|---|---|---|---|---|
| **Execution** | Phase 5 (Process: 5.6 Calculate weights, 5.7 Calculate aggregates) + Phase 6 (Analyze: 6.1 Prepare draft outputs, 6.2 Validate outputs, 6.3 Interpret & explain, 6.4 Apply disclosure control, 6.5 Finalise outputs) | Production (execution) | Data Analysis (execution) | 2.1.2 Metadata management, 2.1.3 Statistical quality management |
| **Management** | Overarching processes (Quality Management, Metadata Management) | Corporate Support (Quality Management, Data Management, Metadata Management), Capability Management (improvement) | Data Quality, Data Governance, Traceability | 2.1.3 Statistical quality management, 4.1.4 Improve management, 4.1.5 Accountability |

## Applying This to Your NSI

### Maturity Assessment Checklist

Use this checklist to assess your NSI's maturity across all four process types — not just execution.

**Design Maturity**

- ☐ Do we have a documented estimation strategy and accuracy-measurement approach per output?
- ☐ Do we have a formal weighting/calibration methodology with defined auxiliary variables and totals?
- ☐ Do we have documented aggregation and model specifications with required diagnostics?
- ☐ Do we have a validation policy and a statistical disclosure control policy defined before estimation begins?
- ☐ Do we have a quality-measurement and output-versioning policy?

**Implementation Maturity**

- ☐ Are our weighting/estimation engines built from design specifications (not ad-hoc scripts)?
- ☐ Does variance estimation reflect both the survey design and imputation uncertainty?
- ☐ Can an output be reproduced identically from the processed input plus the recorded methodology?
- ☐ Is disclosure-control tooling implemented and applied before handover?
- ☐ Do we have a formal testing process (technical, reproducibility, benchmark, accuracy)?

**Execution Maturity**

- ☐ Do we confirm the pre-approved approach and current calibration totals each cycle?
- ☐ Do we attach methodology, weights, and an accuracy measure to every estimate?
- ☐ Do we document model assumptions, diagnostics, and limitations?
- ☐ Do we validate soundness and coherence and explain movements/revisions?
- ☐ Do we apply disclosure control and register every output as an immutable version?

**Management Maturity**

- ☐ Do we track KPIs (accuracy/CV, revisions, reproducibility, timeliness) across cycles?
- ☐ Do we assess coherence and conduct revision analysis?

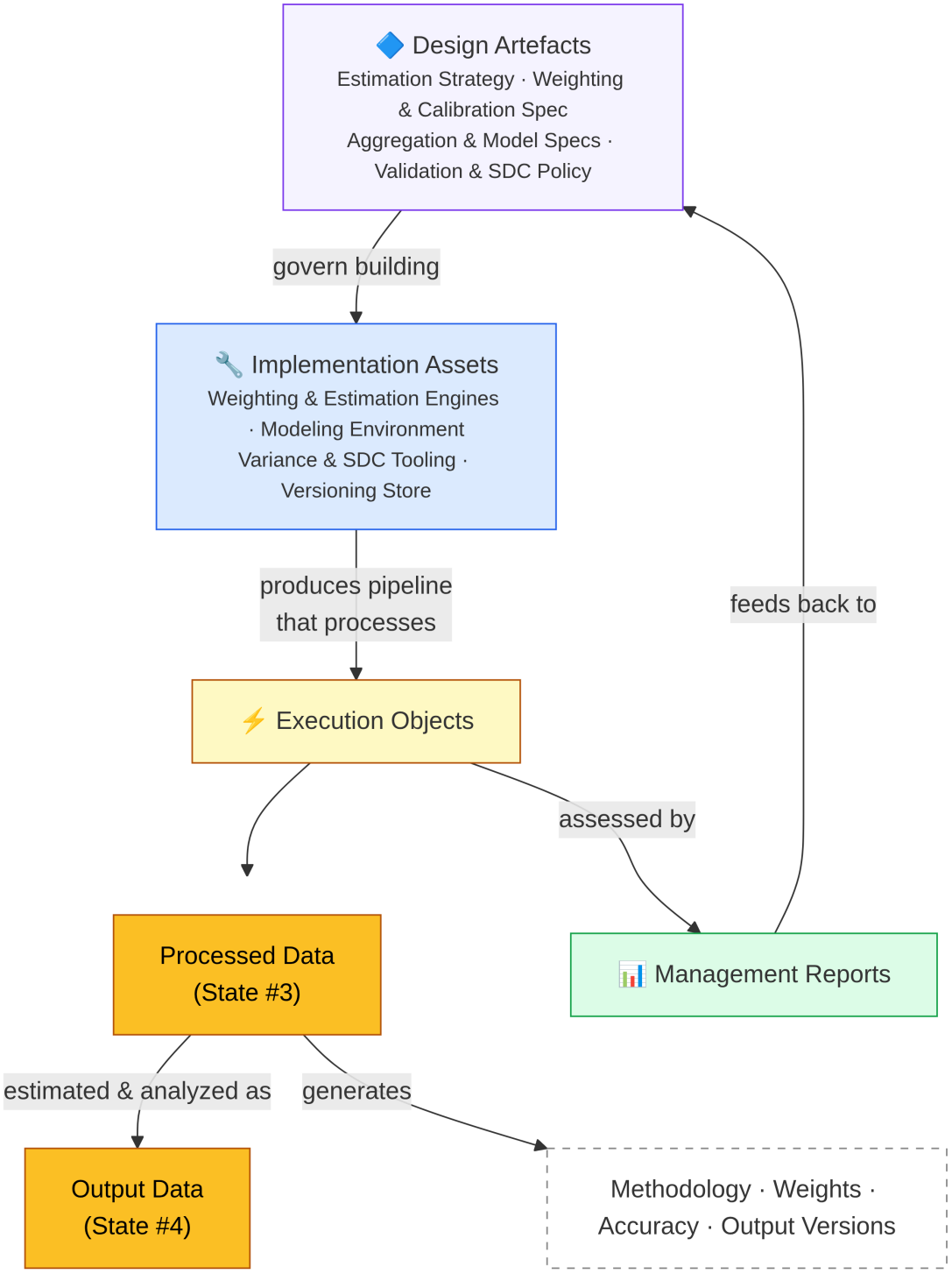



Figure 9: Business Object Lifecycle across all four process types. Design objects govern implementation, which produces the pipeline that estimates and analyses execution objects, which are assessed by management objects, which feed back to design.




Figure 10: Event-Driven Choreography between the upstream domain (VS-03.0) and NSI teams, showing events across the Design → Implement → Execute → Manage cycle. The management phase closes the loop by feeding improvements back to design.

- ☐ Do we govern methodology/model versions and the output-version store?
- ☐ Do we produce reproducibility-audit, disclosure-control, and methodology-compliance reports?
- ☐ Do we have a formal improvement process that feeds back into design?

#### End-to-End Scenarios

Each scenario walks through **all four process types**, demonstrating how the governance pattern works in practice.

#### Scenario A: Labour Force Survey (Design-Based Estimation)

**Design Phase:** Estimation methodologists design a design-based strategy for the employment rate, calibrated to population totals by age × sex × region, with a variance estimator that accounts for the survey design and for the income/non-response imputation flagged in VS-03. Target parameters, domains (national and NUTS-2), coherence criteria with prior quarters, and the disclosure-control policy for small cells are specified.

**Implementation Phase:** Engineers configure the GREG calibration engine (pinned register totals), the estimation engine, and a replication-based variance component that inflates variance for imputed values; they implement coherence checks against prior quarters and small-cell suppression. Reproducibility and back-series tests pass; accuracy measures match expectations on test data.

**Execution Phase:** For Q3-2026, a Processed Data Registered event arrives. The approach and calibration totals are confirmed; calibration converges and weighted margins reproduce the register. The employment rate is estimated at 71.4% (95% CI ±0.5pp national, wider at NUTS-2), with variance inflated for imputation. Coherence checks against Q2 pass; small NUTS-2 cells are suppressed by SDC. An output version is created with methodology, weights, and accuracy, and registered; VS-05.0 is notified.

**Management Phase:** Average CV for key estimates is stable, the coherence-check pass rate is 100%, and the preliminary-vs-final revision is within tolerance. Reproducibility audit confirms the figures regenerate from inputs + methodology. Improvement recommendations: add an auxiliary variable to the calibration to reduce variance for youth-employment estimates. Recommendations fed back to design.

#### Scenario B: Tax Register to Business Statistics (Register-Based + Small-Area Estimation)

**Design Phase:** The team designs register-based aggregation of turnover and employment by NACE × region, and a small-area model for domains where direct aggregates are unstable. Accuracy measures, coherence criteria with national accounts inputs, and disclosure control for dominant-unit cells are specified.

**Implementation Phase:** Engineers build the aggregation pipeline (designed domains, units, rounding), the small-area model with its diagnostics, and dominance-rule suppression for confidentiality. Reproducibility and benchmark tests against the prior year pass; the small-area model's prediction intervals are verified.

**Execution Phase:** Q3-2026: a Processed Data Registered event arrives for the integrated business dataset. Aggregates are computed by NACE × region; unstable small domains are estimated with the small-area model and flagged distinctly from direct aggregates. Dominant-unit cells are suppressed by SDC. Coherence

with national-accounts inputs is checked. `OD-BUS-2026Q3-001` is created with methodology and accuracy, linked to `PD-BUS-2026Q3-001`, and VS-05.0 is notified.

**Management Phase:** Coherence with national accounts is within tolerance; revision analysis shows stable estimates; the small-area model fit is monitored across cycles. Disclosure-control report confirms no dominant-unit disclosure. Improvement recommendation: recalibrate the small-area model after two more cycles of data. Recommendations fed back to design.

#### Scenario C: Web-Scraped Price Data (Index Estimation)

**Design Phase:** The team designs the price-index estimation (e.g., a GEKS or Törnqvist multilateral index) over the COICOP items coded in VS-03, with quality-adjustment treatment, an approach for new/disappearing products, and coherence criteria against the official CPI. The disclosure policy is light (aggregated indices), and the accuracy approach for the index is specified.

**Implementation Phase:** Engineers build the index estimator, the new/disappearing-product handling, and seasonal/quality adjustment, plus coherence checks against the official CPI. Reproducibility and back-series tests pass; the index is benchmarked against the official series on historical data.

**Execution Phase:** For the monthly run, a Processed Data Registered event arrives. The multilateral index is computed over COICOP items using the carry-forward-imputed daily prices flagged in VS-03 (their uncertainty noted), new products are chained in, and the index is validated for coherence against the official CPI movement. An output version with methodology and accuracy is created and registered; VS-05.0 is notified.

**Management Phase:** The scraped index tracks the official CPI within the agreed tolerance; revision magnitude is small. Reproducibility audit passes. Improvement recommendations: refine the new-product chaining rule for one volatile category, and evaluate a longer multilateral window. Recommendations fed back to design.